MCU

File: MCU.kicad_sch

Analog

File: Analog.kicad_sch

Power

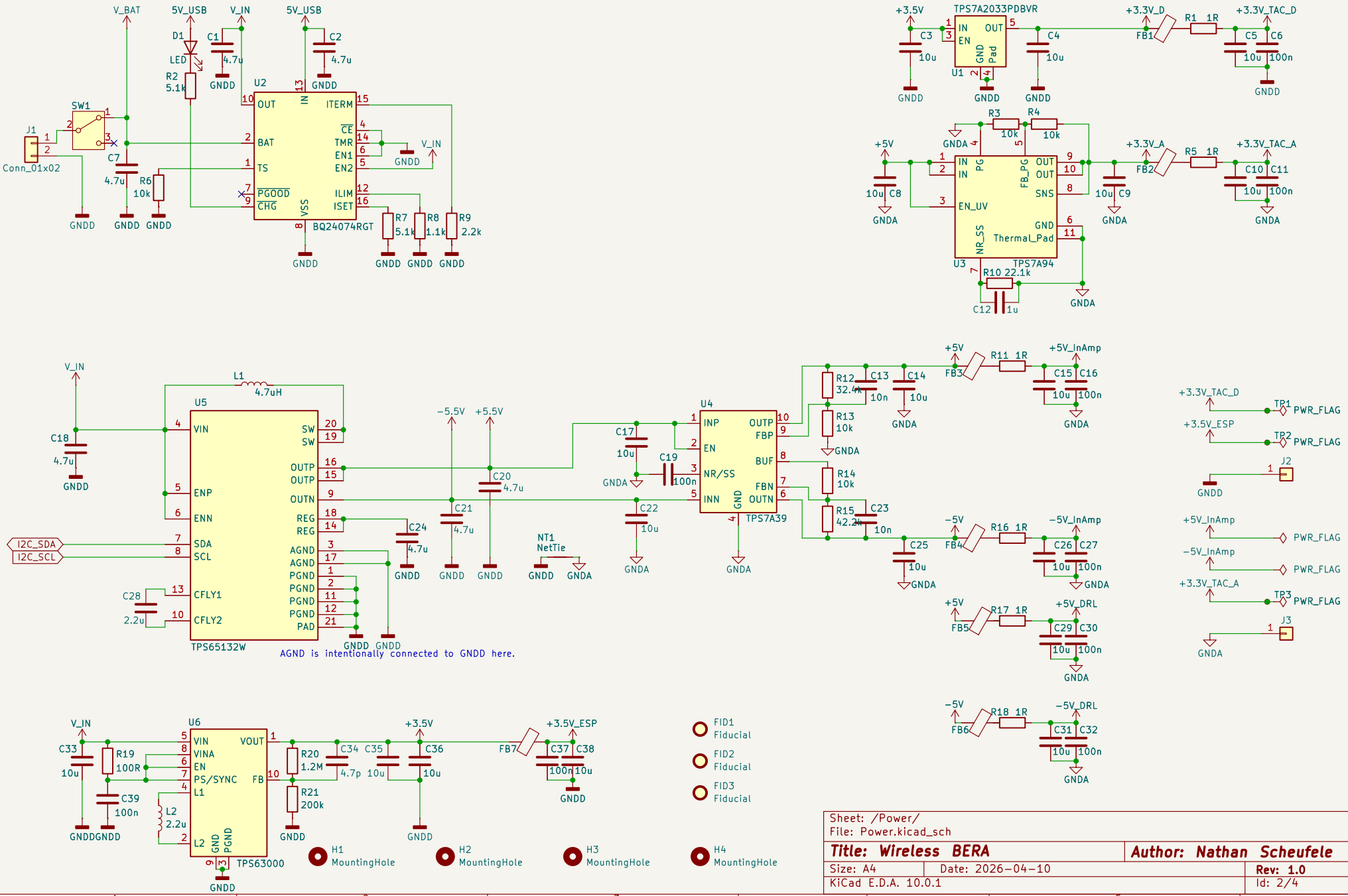
File: Power.kicad_sch

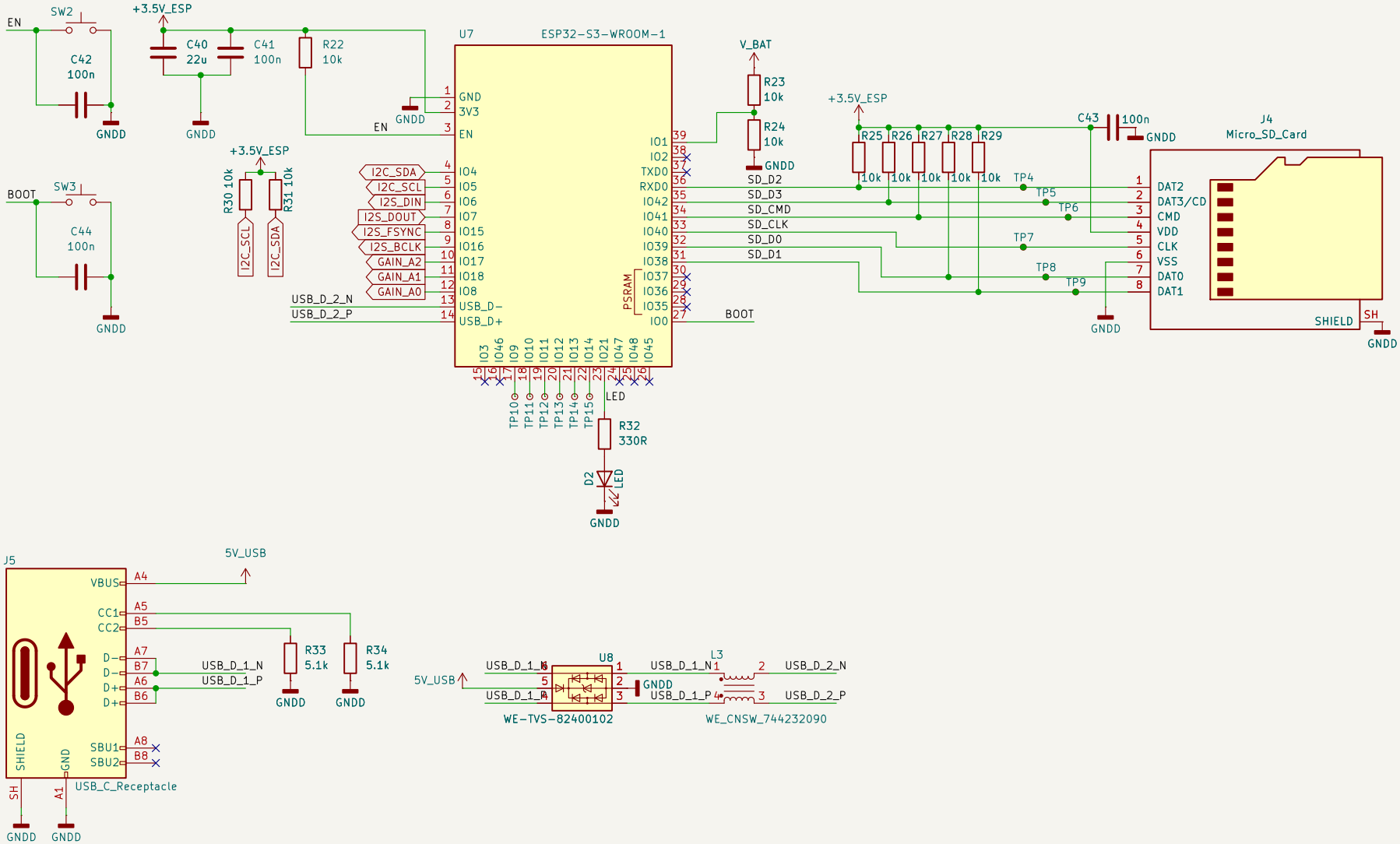
Sheet: /
File: WirelessBERA.kicad_sch

Title: Wireless BERA | **Author: Nathan Scheufele**

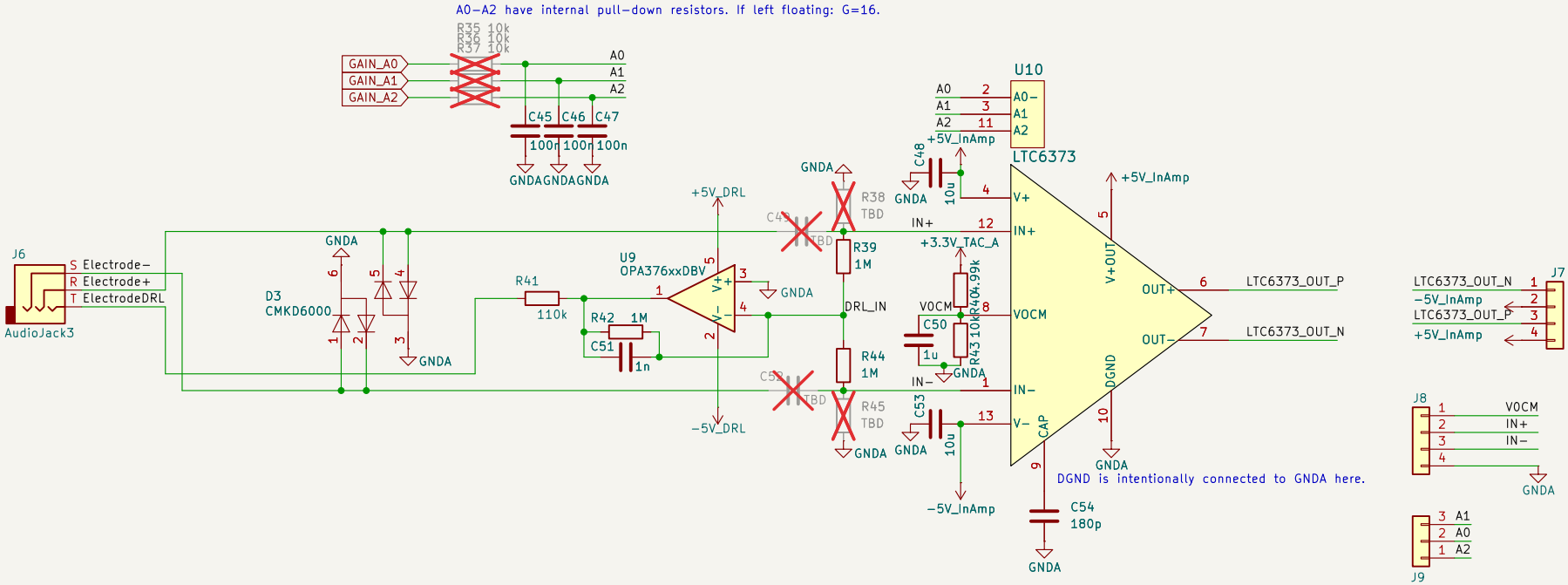
Size: A4 | Date: 2026-04-10 | **Rev: 1.0**

KiCad E.D.A. 10.0.1 | Id: 1/4


V_BAT
5V_USB
V_IN
5V_USB
D1
LED
C1
4.7u
C2
4.7u
R2
5.1k
GNDD
U2
SW1
J1
Conn_01x02
C7
4.7u
R6
10k
OUT
IN
ITERM
BAT
CE
TMR
EN1
EN2
TS
PGOOD
CHG
ILIM
ISET
VSS
BQ24074RGT
R7
5.1k
R8
1.1k
R9
2.2k
+3.5V
TPS7A2033PDBVR
C3
10u
C4
10u
U1
IN
OUT
EN
GND
Pad
+3.3V_D
FB1
R1
1R
+3.3V_TAC_D
C5
10u
C6
100n
R3
10k
R4
10k
+5V
GNDA
C8
10u
C9
10u
U3
IN
IN
PG
FB_PG
OUT
OUT
SNS
EN_UV
NR_SS
GND
Thermal_Pad
TPS7A94
R10
22.1k
C12
1u
+3.3V_A
FB2
R5
1R
+3.3V_TAC_A
C10
10u
C11
100n
V_IN
L1
4.7uH
U5
C18
4.7u
VIN
SW
SW
OUTP
OUTP
OUTN
ENP
ENN
REG
REG
AGND
AGND
PGND
PGND
PGND
PGND
PAD
SDA
SCL
CFLY1
CFLY2
C28
2.2u
TPS65132W
I2C_SDA
I2C_SCL
AGND is intentionally connected to GNDD here.
−5.5V
+5.5V
C20
4.7u
C21
4.7u
C24
4.7u
NT1
NetTie
C17
10u
C19
100n
C22
10u
U4
INP
FBP
EN
BUF
NR/SS
FBN
INN
OUTP
OUTN
GND
TPS7A39
R12
32.4k
R13
10k
R14
10k
R15
42.2k
C13
10n
C14
10u
C23
10n
C25
10u
+5V
FB3
R11
1R
+5V_InAmp
C15
10u
C16
100n
−5V
FB4
R16
1R
−5V_InAmp
C26
10u
C27
100n
+5V
FB5
R17
1R
+5V_DRL
C29
10u
C30
100n
−5V
FB6
R18
1R
−5V_DRL
C31
10u
C32
100n
+3.3V_TAC_D
TP1
PWR_FLAG
+3.3V_ESP
TP2
PWR_FLAG
J2
+5V_InAmp
PWR_FLAG
−5V_InAmp
PWR_FLAG
+3.3V_TAC_A
TP3
PWR_FLAG
J3
V_IN
C33
10u
R19
100R
C39
100n
L2
2.2u
U6
VIN
VINA
EN
PS/SYNC
L1
L2
GND
PGND
VOUT
FB
TPS63000
R20
1.2M
C34
4.7p
C35
10u
R21
200k
+3.5V
C36
10u
FB7
+3.5V_ESP
C37
100n
C38
10u
FID1
Fiducial
FID2
Fiducial
FID3
Fiducial
H1
MountingHole
H2
MountingHole
H3
MountingHole
H4
MountingHole
Sheet: /Power/
File: Power.kicad_sch
Title: Wireless BERA
Author: Nathan Scheufele
Size: A4
Date: 2026-04-10
Rev: 1.0
KiCad E.D.A. 10.0.1
Id: 2/4



Sheet: /MCU/		
File: MCU.kicad_sch		
Title: Wireless BERA		**Author: Nathan Scheufele**
Size: A4	Date: 2026-04-10	**Rev: 1.0**
KiCad E.D.A. 10.0.1		Id: 3/4



Sheet: /Analog/		
File: Analog.kicad_sch		
Title: Wireless BERA		**Author: Nathan Scheufele**
Size: A4	Date: 2026-04-10	**Rev: 1.0**
KiCad E.D.A. 10.0.1		Id: 4/4